CAMBIUM ACADEMY · COURSE 03 · V2

Research with AI

The complete 2026 toolkit. Every instrument, every failure mode, every check.

July 2026 edition. Free forever, no signup. Certificate included when you pass.

WELCOME

Built for people who work with evidence

- **Researchers, grad students, analysts, and the evidence-curious**

  If your work rests on what is true, this course is for you.

- **No code required, code welcomed**

  Every module works without programming. The analysis module goes deeper if you bring R or Python.

- **A browser, about three hours, and $0**

  Self-paced, progress saves in your browser, certificate at the end.

- **Made for one realization**

  Most researchers use one chatbot like a better Google. In 2026 there is an entire instrument rack, and every instrument fails differently.

OUTCOMES

What you will walk away with

- **Send agents to read the literature while you sleep**
  And audit their reports so you never inherit their mistakes.
- **Design studies with a tireless methods consultant**
  Power, sample size, test choice, confounds, before data exists.
- **Turn plain English into working analysis**
  R, Python, SQL, MATLAB, plus transcription and coding for qualitative work.
- **Give your papers eyes, ears, and a voice**
  Machine-read figures and field notes; a library that answers questions and talks like a podcast.
- **Write, translate, and pre-review like a pro**
  With the 2026 disclosure rules of Science, Nature, Elsevier, and NIH actually straight.
- **Leave with your personal stack**
  One page: your instrument for every job, free tiers marked, red lines written down.

YOUR PATH

Five steps, one certificate

1 Read the lecture slides
10 modules, about 45 min

2 Clear the 24 flashcards
Quick recall drills

3 Win the 5 AI Lab games
Instruments, prompts, and fakes

4 Pass the quiz
20 questions, 14 to pass

5 Print your certificate
Instantly, ready for LinkedIn

One tip: keep your own current research project in mind the whole way through. Every module ends with a try-it aimed at YOUR work, not a toy example.

MODULE 1 OF 10

The 2026 research stack

One chatbot is not a research strategy.

What an autonomous AI scientist does in a day

1,500

papers processed in a single run of Kosmos, an autonomous research agent

42,000

lines of analysis code written and executed in that same run

~80%

of its findings its own makers estimate hold up under expert review

1 in 5

findings wrong. That is the bargain: months of work in a day, and you are the audit

Dated fact: Kosmos (Edison Scientific, spun out of the FutureHouse lab), 2026. The point is not this one system. The point is the shape of the deal every AI instrument now offers you.

Seven instruments, one lab

Discover
Agents that search, read, and synthesize the literature.

Design
A methods consultant for power, tests, and confounds.

Analyze
Plain English in, working R/Python/SQL/MATLAB out.

See and hear
Machine eyes for figures and notebooks; a library that talks.

Write
Drafting, translation, grants, and pre-review.

Reproduce and disclose
Documentation, real references, and the paper trail.

Each module hands you one instrument, one measured 2026 fact about where it fails, and the one check that makes it safe.

The sentence that governs this course

> *"AI supplies speed. You supply trust."*

- **Speed is now nearly free**
  Reports in minutes, code in seconds, transcription in real time.
- **Trust was never free**
  Verification is a skill, and it is the one this certificate actually certifies.
- **The rack changes monthly, the rule does not**
  Tools rotate. Judgment compounds.

If you keep one sentence from the next three hours, keep this one.

Sort your own week onto the rack

- **List your last week's research tasks**
  Reading, planning, analyzing, writing, formatting, checking.

- **Tag each with its instrument**
  Discover, design, analyze, see-hear, write, reproduce, disclose.

- **Find the two you still do fully by hand**
  Those two are where this course pays for itself first.

Keep the list. Module 10 turns it into your personal stack, and that page is your capstone.

Prompting for researchers

The foundation everything else uses. So it comes first.

Context. Deliverable. Format. Uncertainty.

- **Context: what it cannot guess**
  Your field, your data, your constraints, what you already tried, what good looks like.

- **Deliverable: a job, not a topic**
  "Draft a limitations paragraph for THIS design" beats "tell me about limitations."

- **Format: the exact shape**
  Word limit, structure, citation style, table columns. Cheap to say, transformative to get.

- **Uncertainty: demanded, every time**
  One standing sentence: "Flag what you are not sure of and what I must verify."

The fourth line is the researcher's upgrade. It converts a confident narrator into an honest assistant, and it costs you nothing.

Demand uncertainty like a colleague would

“"Flag what you are not sure of, state the assumptions you just made, and tell me what I should verify before relying on this."”

- **Why it works**

  Models are tuned to be helpful and fluent. An empty answer feels like failure to them, so they fill gaps. This sentence gives permission to show the seams.

- **What you get back**

  Hedges where hedges belong, assumptions surfaced, and a verification list you would have had to invent yourself.

Paste it into any factual or analytical request for the rest of your career.

Four prompts that earn their keep

- **"Act as reviewer 2"**
  Attack this abstract: clarity, overclaims, missing limitations. Tireless and shameless, in private.
- **"List the assumptions you just made"**
  Run after any analysis or summary. The list is where the errors live.
- **"Give the opposing interpretation"**
  Forces the model off your side of the argument, which is where you are blindest.
- **"What would falsify this?"**
  The most scientific sentence you can say to a machine.

Notice none of these are magic words. They are the questions a good committee asks, made cheap.

Roles shape tone, not truth

- **"You are a world-class statistician" changes the costume**
  Style, vocabulary, confidence: yes. Factual accuracy: measured slightly WORSE in testing.
- **For facts and stats, ask plainly**
  Plain question, full context, demanded uncertainty.
- **Save personas for voice**
  "Explain this to a first-year cohort" is a legitimate style instruction.

Carried from Course 02, where the 2026 evidence is walked through slide by slide.

Resurrect your worst AI answer

- **Find a recent AI answer that disappointed you**

  Everyone has one from this week.

- **Rerun it with the recipe**

  Context, exact deliverable, format, and the standing uncertainty sentence.

- **Diff the two answers**

  The delta is the skill. It typically takes under two minutes to apply.

MODULE 3 OF 10

Discover: the literature pipeline

Search. Map. Send the agents. Keep it alive.

Four gears, lowest to highest

- **Gear 1: search-grounded Q&A**
  A quick cited answer with clickable receipts. Minutes of your attention.
- **Gear 2: citation maps**
  See a field's neighborhood around one seed paper. Coffee-break scale.
- **Gear 3: deep research agents**
  A synthesized, cited report from dozens of sources. Run it and walk away.
- **Gear 4: standing alerts**
  The review that keeps updating itself after you stop looking.

Most researchers only ever use gear 1, and use it like a search engine. The next slides hand you the other three.

Citation mapping tools (July 2026)

Connected Papers
Drop in one seed paper, get a visual similarity graph of its neighborhood, including papers that never cite each other.

ResearchRabbit
Chain recommendations interactively: semantic similarity plus citations, with alerts as collections grow.

Litmaps
A living map over 260M+ papers (Crossref, Semantic Scholar, OpenAlex), sorted by date and citations.

The workflow: one good seed paper in, one afternoon of reading list out. The map shows you the adjacent method community you did not know existed.

Deep research agents, measured

50-200

sources per run, OpenAI Deep Research (10 to 30 minutes)

30-150

sources per run, Gemini Deep Research (5 to 15 minutes)

20-100

sources, Claude Research: fewer sources, deeper synthesis per source

Free tier

exists: Perplexity runs limited deep research runs at $0

Every major AI now ships one. You describe the question; it plans, searches, reads, cross-references, and writes a cited report while you do something better with your time.

Why Elicit and Undermind beat the generalists here

- **They search closed, verified paper databases**

  Not the open web. The corpus is published research, so citations resolve by construction.

- **They speak researcher**

  Screening columns, effect extraction, PRISMA-style workflows, exportable tables.

- **The generalists still matter**

  For breadth beyond papers: policy, preprints, datasets, news, code.

Rule of thumb: literature review means academic agent first, generalist second. Everything else means the reverse.

A first draft with leads

“Treat every agent report as a smart first draft assembled by a fast assistant who occasionally pads the bibliography.”

- **The polish is the trap**

  Clean sections and numbered references SIGNAL audited. Nothing was audited.

- **The audit is one habit**

  Find the 2 or 3 sources the conclusion actually rests on. Three-click each one.

- **Then set the alert**

  Turn the finished question into a standing search so the answer stays current.

Carried from the trust module ahead, because it is the single habit that makes gear 3 safe.

The two-agent shootout

- **Pick a question from your actual research**
  Narrow beats broad for this test.
- **Run it through one generalist agent and Elicit**
  Same wording, both barrels.
- **Count sources that survive a click**
  Existence, correct venue, actually says it. The difference IS the lesson.

MODULE 4 OF 10

Trust: the citation trap

The same machine that finds truth at scale manufactures the appearance of truth at scale.

Fabrication, measured at scale

1 in 277

papers contained a fabricated reference by early 2026, up ~6x in three years (Lancet analysis, 2M+ papers, 97M citations)

60%+

of news-citation queries answered WRONG by AI search tools in Columbia Journalism Review's test, in a fully confident tone

18-28%

of citations fabricated by frontier models on literature-review tasks in 2026 benchmarks

3 clicks

is all it takes to catch essentially every fake: exists, matches, says it

These four numbers are why "the AI said so" is never a final answer in this course.

Citation-shaped text

- **Models learned what a citation looks like**
  Plausible authors, real-sounding journal, well-formed DOI, sensible pages.
- **When unsure, they complete the pattern**
  No lookup happens. A convincing forgery fills the slot.
- **So the glance is worthless**
  The fakes are optimized, by their training, to survive exactly the check most people apply.

Nobody programmed deception. Plausible text is the product, and that is precisely the problem.

Read across, not down

S Stop
Do you actually know this source and this claim?

I Investigate the source
New tab. Who is behind this? Thirty seconds.

F Find better coverage
Search the claim itself. Do independent sources report it?

T Trace to the original
Follow the stat upstream. Context changes everything.

Lateral reading: leave the page and let independent sources judge it. Faster AND more reliable than staring harder.

The only rule this course carves

> *“A citation you never opened is a citation you cannot trust.”*

- **Click 1: does it exist?**
  Scholar or the DOI. No result is your answer.
- **Click 2: is it what it claims?**
  Right authors, right venue, right year. Fakes borrow real titles with wrong details.
- **Click 3: does it say what the AI said?**
  Open the abstract. Misattribution is the subtlest fake of all.

Thirty seconds per citation that matters. The AI Lab's Citation Hunt makes this a reflex, with score and streaks.

MODULE 4 · TRY IT NOW

Play the Lab, then hunt in the wild

- **Win Citation Hunt, Lateral Race, and Receipts-or-Vibes**
  The Lab's first three games train exactly this module.
- **Then ask any chatbot for three sources on a niche topic you know**
  Fakes cluster where training data is thin.
- **Open all three**
  Many researchers meet their first fabricated citation inside five minutes. It changes how you read forever.

MODULE 5 OF 10

Design before data

The cheapest AI hour in your whole project, and the one nobody uses.

Consult the machine before the data exists

- **Power and sample size, sanity-checked**
  Before you commit a season of fieldwork to an underpowered design.
- **Test selection with assumptions stated**
  "Which analysis fits this design, and what must be true for it to be valid?"
- **Confound hunting**
  "List the confounds a hostile reviewer would raise for this design."
- **Pre-registration drafting**
  Hypotheses, exclusions, and the analysis plan, written down before temptation exists.

Errors here are silent and permanent. No amount of clever analysis rescues a broken design, which is why this hour is the cheapest one.

Show the math or it did not happen

“"Give me the sample size, AND the formula, AND every assumption behind it, then say what would change if each assumption is wrong."”

- **Numbers without derivations are vibes**
  A power calculation you cannot inspect is a guess wearing a lab coat.
- **Cross-check with a second tool**
  Two independent calculations that agree are worth ten confident ones.
- **Then take it to a human when stakes are high**
  AI is the tireless first consult, not the sign-off.

The statistics remain yours. "The AI chose the test" is not a methods section.

Stress-test your next study in ten minutes

- **Describe the study in six sentences**
  Question, design, sample, measures, analysis, constraint.
- **Ask for the three most likely fatal flaws**
  And the sample size math, shown in full.
- **Ask what a hostile reviewer would propose instead**
  You do not have to agree. You have to have heard it before they say it.

MODULE 6 OF 10

Analyze: your AI data analyst

Plain English in. Working analysis out. Your judgment in between.

What 2026 assistants actually see and do

- **Posit Assistant sees your dataframe, your code, AND your rendered plot**
  Purpose-built for data work in Positron and RStudio, reasoning across R and Python in one analysis.
- **Agentic assistants navigate whole repositories**
  Claude Code and Copilot agents read, run, and revise multi-file analyses.
- **Browser analysts run plain-English analyses**
  Julius and ChatGPT execute code on your uploaded data and hand back plots.
- **MATLAB ships its own AI chat**
  The big proprietary stacks are in the same race.

The interface stopped being autocomplete and became a colleague who can see your screen. The safety rules change accordingly.

Five steps that keep you the analyst

- **Describe data + goal + constraints**
  Column meanings, units, what a row is, what success looks like.
- **Require commented code**
  Every block explains itself, in your language of choice.
- **The explain-it-back rule**
  AI walks through every line before you run anything.
- **The tiny-test rule**
  Run on ten rows you can verify by hand. Then, and only then, scale.
- **Keep the transcript**
  The conversation IS your analysis log. Save it with the code.

Where it silently fails: plausible-but-wrong test choices, hallucinated function arguments, silent unit and join errors. Every failure mode above is caught by the loop's steps 3 and 4.

Half of research is not a spreadsheet

- **Transcription is effectively solved**

  Whisper-class models turn interview hours into text in minutes, including accents and crosstalk that used to cost transcription budgets.

- **Thematic coding: AI proposes, you adjudicate**

  First-pass themes and codings at scale, with every decision logged. The interpretive authority stays human.

- **Survey open-ends at scale**

  Thousands of free-text responses, clustered and quantified in an afternoon.

Keep an audit trail of accepted and rejected codings. "The AI coded it" is not a methodology; "AI-assisted coding with human adjudication, trail attached" is.

The messy CSV challenge

- **Hand an assistant your messiest real CSV**
  With a plain-English description of what each column means.
- **Ask for cleaning plus one labeled plot**
  In R or Python, every step commented, explained back before running.
- **Run the tiny test first**
  Ten rows, hand-checked. Then the full file.

MODULE 7 OF 10

See and hear

Machine eyes for your figures and field notes. A voice for your paper library.

Multimodal AI for research material

- **Extract numbers from published charts**
  Digitize the figure from that 1998 paper nobody has the data for.
- **First-pass reads of micrographs and maps**
  Descriptions, counts, anomalies, at triage speed.
- **Transcribe handwritten field notebooks**
  Including the writing you can barely read yourself.
- **Pull tables out of photos and PDFs**
  The eternal scanned-appendix problem, mostly solved.

The measured limit: vision models still misidentify fine-grained structures in cluttered scientific imagery that trained experts catch. Brilliant interns with astigmatism: fast, useful, and never the final reader.

Hypothesis, then spot-check

- **Every extracted number gets spot-verified**
  Check a sample against the source before any extracted dataset enters your analysis.
- **Every identification is a hypothesis**
  A model calling a structure on a micrograph is a lead for the expert, never a result for the paper.

Used this way, machine eyes are a triage superpower with zero downside. Skipped, they are a retraction generator.

Source-grounded assistants (the NotebookLM class)

- **Upload your papers, ask questions, get cited answers**

  Answers ground in YOUR corpus, not in training memory, and cite back to the page.

- **Audio overviews you can interrupt**

  Podcast-style discussions of your sources, in formats from brief to critical debate, that take your questions mid-conversation.

- **Maps, decks, and quizzes on demand**

  The library becomes study materials while you commute.

NotebookLM is the flagship demo; the CAPABILITY, source-grounded Q&A over your own corpus, is the thing to learn. Products rotate.

Containment, not elimination

“A source-grounded answer can still be wrong, but it is wrong INSIDE a corpus you can check, with citations that point at pages you own.”

- **Open chat free-associates from memory**
  Its errors can come from anywhere, so checking means searching the world.

- **Grounded chat errs inside your uploads**
  Checking means opening the cited page. The error surface shrinks by orders of magnitude.

This is the same receipts principle from Module 3, applied to your own bookshelf.

One page, one chart, one podcast

- **Photograph a handwritten page and a published chart**
  Ask for transcription and extracted values. Count the errors yourself.
- **Upload three papers from your field**
  Generate a critical-format audio overview.
- **Interrupt it**
  Ask the question you would ask the authors. That moment sells the whole module.

MODULE 8 OF 10

Write: drafts, translation, grants, review

The most-used AI research task on earth, and the most policed.

What writing help looks like when it is honest

- **Revision and clarity passes**

  Tighten, restructure, de-jargon: your ideas, better sentences.

- **Translation and English polishing**

  The great equalizer for non-native speakers, and at many venues it now counts as exempt copy-editing. Check YOUR venue.

- **The unglamorous fleet**

  Abstracts, cover letters, plain-language summaries, response-to-reviewers scaffolds.

- **Grant support**

  Aims critiqued, reviewer simulated, boilerplate drafted. The science stays yours; NIH has signaled AI-substantially-written proposals can be rejected.

The line that never moves: AI assists the writing. You remain the author, and the author answers for every sentence.

The rules now differ FOR REAL

Science
Disclosure of ALL AI use required, in the acknowledgments. Including copy editing.

Nature portfolio
Copy-editing exempt from disclosure; AI-generated images banned; methods-level disclosure for substantive use.

Elsevier
Dedicated declaration section for generative AI in the writing process; no AI authorship.

NIH
Reviewers barred from feeding applications to AI; proposals substantially written by AI can be rejected.

One-size-fits-all advice about AI writing is now wrong by construction. The habit is: check the venue's actual policy page at submission time, every time.

Reviewer 2, on demand, in private

“"Attack this manuscript as a skeptical reviewer: clarity, methods gaps, statistical red flags, overclaimed conclusions, missing limitations. Be specific and merciless."”

- **It is tireless and shameless**

  Exactly what you want BEFORE humans see it.

- **It is already on the other side too**

  Roughly 21% of reviews at a major AI conference were flagged as substantially AI-written, and ~12% of review text at a major journal family.

- **So assume a hybrid audience**

  Write for humans, and expect machines to read you first.

Where writing help becomes misconduct

- **Never upload someone ELSE'S unpublished work**
  Manuscripts and grants you review are confidential. NIH bars reviewers from feeding applications to AI, full stop.
- **Never let AI invent citations, data, or quotes**
  Module 4 taught you exactly how convincingly it will.
- **Never skip the venue's declaration**
  Disclosure costs a sentence. Discovery costs a reputation.

Try it tonight: paste YOUR abstract, ask for the three harshest justified referee comments, and fix one before bed.

MODULE 9 OF 10

Reproducible and intact

Analyses anyone can rerun. References anyone can open.

And the assistant that fixes both, if asked

- **Crisis one: analyses nobody can rerun**

  Undocumented scripts, unpinned environments, the folder called final_v2_REAL.

- **Crisis two: reference lists nobody checked**

  Module 4 measured this one: 1 in 277 papers and rising.

- **The fix costs one sentence each**

  The same assistant that wrote your analysis will write the README, pin the environment, and document the workflow. Most people never ask.

Reproducibility went from a virtue to a prompt. "Document this script, list dependencies with versions, and write a README a stranger could follow" takes eleven seconds to type.

Four prompts for your existing code

- **"Document every function and non-obvious line"**
  Comments a stranger, including future-you, can follow.
- **"Pin this environment"**
  requirements.txt, renv, or a container recipe, with versions.
- **"Turn this session into a script"**
  Interactive spelunking becomes a rerunnable pipeline.
- **"Write the README"**
  Data expected, steps to run, outputs produced, one page.

AI drafts, you correct. The audit trail from Module 6's safe loop completes the package.

The two nevers

“Never hand-type a citation. And never trust an AI-typed one.”

- **Import from the source of record**
  Zotero pulls metadata from Crossref and publishers; Better BibTeX keeps keys stable.
- **AI plugins are a reading layer, not a source**
  Beaver and PapersGPT-class tools chat over your library; the BIBLIOGRAPHIC RECORD still comes from the registry.
- **The final gate: DOIs resolve**
  One pass before submission: every DOI opens, every entry matches. Minutes, not hours.

BibTeX generated from a model's memory is Module 4's fabrication problem wearing a bow tie.

Two checks on your current project

- **Have AI document one existing script**
  Then correct its two wrong guesses. That correction IS the documentation working.
- **Run your reference list through a DOI check**
  Count the casualties. Almost every list has at least one.

MODULE 10 OF 10

Responsible AI, and your personal stack

The capstone: judgment, written down.

Disclosure, measured

~70%

of journals now have an explicit AI policy

~0.1%

of 75,000 recent papers actually disclosed AI use

1 sentence

is what proper disclosure usually costs

4 rules

Find well. Verify well. Cite well. Disclose well.

That gap will close, and it will close on the disclosers' terms. Be early.

Five lines to work by

- **Verify before you rely**

  Every load-bearing claim, number, and citation. You know all the checks now.

- **Disclose per the venue's ACTUAL rule**

  Module 8's table, checked fresh at submission.

- **Guard sensitive data absolutely**

  Confidential, pre-publication, human-subjects data never touches a cloud chatbot. Local and institution-hosted models exist exactly for this.

- **Watch the bias**

  Training data shapes answers: languages, regions, and fields are not equally represented. Ask what is missing.

- **You remain the accountable author**

  Every instrument in this course assists. None of them signs.

The sensitive-data line is hard. If your data could not go in an email to a stranger, it does not go in a consumer chatbot. When in doubt: local model, institutional deployment, or synthetic stand-in data.

Build your personal stack, one page

1 Discover
Your agent + your map tool + one alert

2 Design + analyze
Your methods consult + your coding assistant

3 See, hear, write
Your vision tool, library assistant, and writing setup

4 Reproduce
Your documentation habit + reference pipeline

5 Red lines
Your venue's disclosure rule + your data rules, written down

6 Free tiers marked
So the stack survives a $0 month

This page is the deliverable. Tools on it will rotate; the shape of it will not. The certificate says you can pick instruments with judgment, not that you memorized this month's names.

THE POCKET CHECKLIST

Screenshot this

- **Before you ask**

  Full context, exact deliverable, format, demanded uncertainty.

- **Before you trust**

  Load-bearing claims verified laterally; citations opened: exists, matches, says it.

- **Before you run**

  Code explained back, tiny test passed, transcript saved.

- **Before you submit**

  References resolve, environment pinned, venue's AI policy checked, disclosure written.

- **Always**

  AI supplies speed. You supply trust.

Where to go deeper, all free

- **The course resources page**

  Every tool and study from these slides, linked and dated, re-verified at release.

- **Your own two-agent shootout**

  Module 3's try-it is the gift that keeps teaching: rerun it quarterly as tools evolve.

- **The Academy path**

  Prompting Essentials covers the asking skill in depth; the local-models course on the roadmap covers the sensitive-data lane hands-on.

Everything on the resources page was verified by hand this month.

YOUR TURN

Find well. Verify well. Cite well. Disclose well.

- **Step 1, the lecture: done**
  You just finished all ten instruments.
- **Step 2: the 24 flashcards**
  Quick, and they recycle what you miss.
- **Step 3: win all 5 AI Lab games**
  Citation Hunt, Lateral Race, Receipts-or-Vibes, Pick Your Instrument, Prompt Surgery.
- **Then the quiz**
  20 questions, 14 to pass, certificate on the spot.